MediXO EduX
AI Academic Intelligence

EXAMINATION
INTELLIGENCE
REPORT

FULL EXAM ANALYSIS REPORT

DETAILED PERFORMANCE • QUESTION ANALYSIS • LEARNING INSIGHTS

JEE Main Mock Test - Phase 2

Combined (Physics, Chemistry, Mathematics) | All Subjects

22.3%
Score

Marks Obtained
67 / 300

23 Correct | 26 Incorrect | 26 Skipped



Student	**SUJJAL KUMAR NATH**
Roll Number	**260094**
Exam Date	**31 Oct 2026**
Total Marks	**300**

Subject Analysis | Chapter Mastery | Question-wise Review | Error & Learning Insights | Revision Plan & Next Steps

DISCIPLINE
CREATIVITY
PROGRESS

ABRIGHTER
TOMORROW
TOGETHER

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

75 questions in the paper



- Correct 23
- Incorrect 26
- Skipped 26

Mistake distribution

Count by published classification

Unit Error	5
Formula Error	6
Knowledge Gap	3
Calculation Error	11
Incomplete Reasoning	1

Source classifications; rows are measured independently.

Subject-wise performance

Accuracy by subject

Physics	28%
Chemistry	44%
Mathematics	20%

Physics 17/100 Chemistry 37/100 Mathematics 13/100

Important performance signals

GRADE **D**	RANK **11**	BATCH RANK **11**
COHORT SIZE **51**	OVERALL SPEED **2.4**	CONFIDENCE INDEX **46.9**
READINESS SCORE **19.7**	HEALTH SCORE **19.7**	NEGATIVE MARKS **25**
GUESS ATTEMPTS **0**	ATTEMPT RATIO **65.3%**	TOTAL QUESTIONS **75**
OVERALL ACCURACY **46.94%**	BADGE **Completed**	

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	17 / 100	28%	12	Medium	Acceleration Calculation · Motion Under Gravity · Acceleration-Time Graph	Kinematics Graphs · Vertical Motion Under Gravity · Consecutive Time Distances
Chemistry	37 / 100	44%	10	Medium	Electronic Configuration · Nodes in Orbitals · Aufbau Principle	Discovery of Subatomic Particles · Orbital Energies and Penetration · Photoelectric Effect and Ionization
Mathematics	13 / 100	20%	11	Medium	Trigonometric Identities · Exponential Equations · Telescoping Sums	Modulus Properties · Cosine Series · Euler Form

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	Laws of Motion & Mechanics	Kinematics · Laws of Motion	Kinematics · Laws of Motion
Chemistry		Structure of Atom	

Strong, weak and critical chapters by subject · continued

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Mathematics		Sets and Relations · Sequences and Series · Trigonometry · Quadratic Equations · Complex Numbers · Inequalities	Sets and Relations · Sequences and Series · Quadratic Equations · Complex Numbers · Inequalities

Chapter mastery · grouped by subject

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Kinematics	35.3%	16	71m 48s	Critical
Laws of Motion	0%	-3	19m 54s	Critical
Laws of Motion & Mechanics	100%	4	48s	Strong
Chemistry				
Structure of Atom	40.9%	30	26m 54s	Weak
Classification of Elements	66.7%	7	3m 36s	Average
Mathematics				
Sets and Relations	0%	0	6m 42s	Critical
Sequences and Series	20%	3	2m 36s	Critical
Trigonometry	50%	6	10m 30s	Weak
Quadratic Equations	33.3%	8	5m 12s	Critical
Complex Numbers	0%	-3	29m 12s	Critical
Inequalities	0%	-1	2m 48s	Critical

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics			
Kinematics	40%	71m 48s	40%
Laws of Motion	11%	19m 54s	11%
Laws of Motion & Mechanics	0%	48s	0%
Chemistry			
Structure of Atom	15%	26m 54s	15%
Classification of Elements	2%	3m 36s	2%
Mathematics			
Sets and Relations	4%	6m 42s	4%
Sequences and Series	1%	2m 36s	1%
Trigonometry	6%	10m 30s	6%
Quadratic Equations	3%	5m 12s	3%
Complex Numbers	16%	29m 12s	16%
Inequalities	2%	2m 48s	2%

Time-management metrics

AVERAGE PER QUESTION
2.4

NAVIGATION COUNT
0

MANAGEMENT SCORE
19.7

FASTEST QUESTION
Q34

FASTEST TIME
0.02

SLOWEST QUESTION
Q1

SLOWEST TIME
29.43

Sectional allocation and efficiency

SECTION	ALLOCATED	USED	EFFICIENCY
Physics	180	92.5	51.4
Chemistry	180	30.5	16.9
Mathematics	180	57.1	31.7

04 QUESTION ANALYSIS

Question-by-question analysis

75 questions reviewed · published explanations reproduced in full

01 Physics

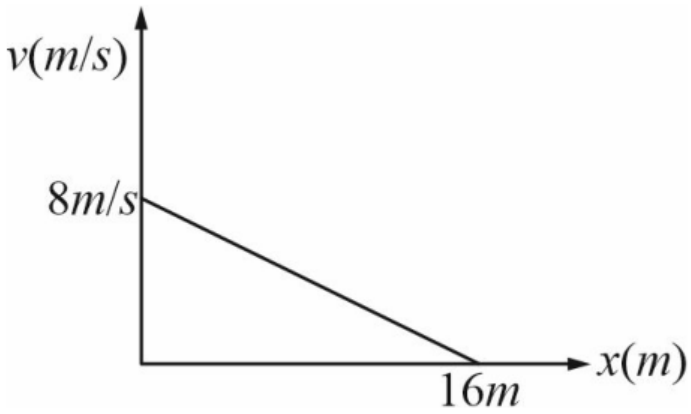
25 questions

QUESTION 01 PHYSICS

INCORRECT

Kinematics · Kinematics Graphs · Medium · MCQ · −1 mark · 29m 26s

The v-x graph of a particle moving along x-axis is as shown in the figure. Then acceleration of the particle at x = 8 m is:



OPTIONS

A. 2 m/s^2

B. -2 m/s^2

C. 4 m/s^2

D. -4 m/s^2

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Unit Error · The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

EXPLANATION

Acceleration a = v (dv/dx). From the graph, slope dv/dx = -8/16 = -1/2. At x = 8 m, v = 8 -(1/2)(8) = 4 m/s. Therefore, a = 4 * (-1/2) = -2 m/s^2.

MISTAKE NOTE

The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

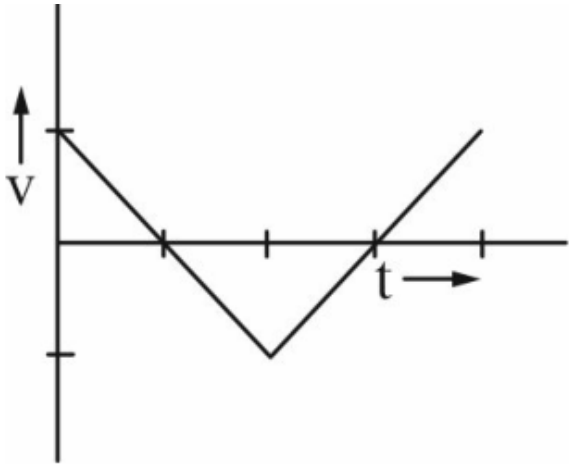
Physics · Q2 · question continued

QUESTION 02 PHYSICS

INCORRECT

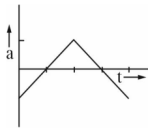
Kinematics · Kinematics Graphs · Medium · MCQ · −1 mark · 2m 47s

The graph shown in figure shows the velocity v versus time t for a body. Which of the graphs represents the corresponding acceleration versus time graphs?

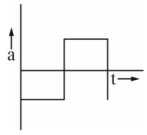


OPTIONS

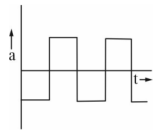
A. (A)



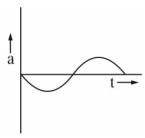
B. (B)



C. (C)



D. (D)



STUDENT RESPONSE

C

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Unit Error · The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

EXPLANATION

The slope of the v-t graph is negative and constant for the first half, and positive and constant over the second half.

Therefore, acceleration is a negative constant followed by a positive constant, corresponding to graph (B).

Physics · Q2 · explanation continued

MISTAKE NOTE

The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

QUESTION 03 PHYSICS

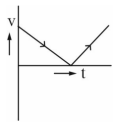
INCORRECT

Kinematics · Vertical Motion Under Gravity · Medium · MCQ · −1 mark · 1m 14s

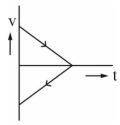
A ball is thrown vertically upwards. Which of the following graphs represents the velocity-time graph of the ball during its flight? (air resistance is neglected)

OPTIONS

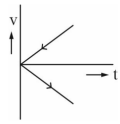
A. (A)



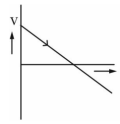
B. (B)



C. (C)



D. (D)



STUDENT RESPONSE

B

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Vertical Motion Under Gravity.

EXPLANATION

v = u - gt. The velocity-time relation is a straight line with a constant negative slope (-g), crossing the time axis when v = 0 at the highest point and continuing downwards, corresponding to graph (D).

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Vertical Motion Under Gravity.

QUESTION 04 PHYSICS

CORRECT

Kinematics · Acceleration Calculation · Easy · MCQ · 4 marks · 50s

If the displacement of a particle is $x = 2t^2 + t + 5$ meters, what will be its acceleration at $t = 5$ seconds?

OPTIONS

A. 21 m/s^2

B. 20 m/s^2

C. 4 m/s^2

D. 10 m/s^2

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

v = dx/dt = 4t + 1 m/s. a = dv/dt = 4 m/s^2. The acceleration is 4 m/s^2 at all times.

QUESTION 05 PHYSICS

CORRECT

Kinematics · Motion Under Gravity · Medium · MCQ · 4 marks · 4m 09s

A man throws balls with the same speed vertically upwards, one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given $g = 9.8$ m/s^2)

OPTIONS

A. At least 0.8 m/s

B. Any speed less than 19.6 m/s

C. Only with speed 19.6 m/s

D. More than 19.6 m/s

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

For more than 2 balls to remain in the sky, the time of flight T of each ball must exceed 4 seconds (2 × 2 s). T = 2u/g > 4 ⇒ u > 2g = 2 × 9.8 = 19.6 m/s.

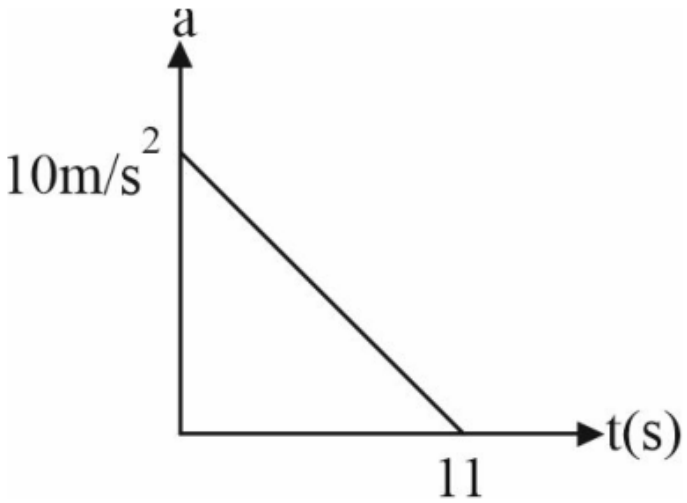
Physics · Q6 · question continued

QUESTION 06 PHYSICS

CORRECT

Kinematics · Acceleration-Time Graph · Medium · MCQ · 4 marks · 2m 57s

A particle starts from rest. Its acceleration (a) versus time (t) graph is as shown in the figure. The maximum speed of the particle will be:



OPTIONS

A. 110 m/s

B. 55 m/s

C. 550 m/s

D. 660 m/s

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

v_max - v_initial = Area under a-t graph = 1/2 × base × height = 1/2 × 11 × 10 = 55 m/s. Since u = 0, v_max = 55 m/s.

QUESTION 07 PHYSICS

INCORRECT

Kinematics · Consecutive Time Distances · Medium · MCQ · −1 mark · 2m 17s

A body travels for 15 seconds starting from rest with constant acceleration. If it travels distances S_1, S_2, and S_3 in the first five seconds, second five seconds, and next five seconds respectively, the relation between S_1, S_2, and S_3 is:

OPTIONS

A. $S_1 = S_2 = S_3$

B. $5S_1 = 3S_2 = S_3$

C. $S_1 = (1/3)\ S_2 = (1/5)\ S_3$

D. $S_1 = (1/5)\ S_2 = (1/3)\ S_3$

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Consecutive Time Distances.

EXPLANATION

For motion from rest under constant acceleration, distances travelled in consecutive equal time intervals are in the ratio of odd numbers: $S_1 : S_2 : S_3 = 1 : 3 : 5$, hence $S_1 = S_2 / 3 = S_3 / 5$.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Consecutive Time Distances.

QUESTION 08 PHYSICS

CORRECT

Kinematics · Distance in nth Second · Medium · MCQ · 4 marks · 1m 15s

A particle is moving in a straight line with initial velocity u and uniform acceleration a. If the sum of the distance travelled in t-th and (t + 1)-th seconds is 100 cm, then its velocity after t seconds, in cm/s, is:

OPTIONS

A. 80

B. 50

C. 20

D. 30

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

$S_t + S_{(t+1)} = [u + a(t - 1/2)] + [u + a(t + 1/2)] = 2(u + at) = 100$ cm. Therefore, velocity after t seconds is $v = u + at = 50$ cm/s.

QUESTION 09 PHYSICS

INCORRECT

Kinematics · Multi-stage Motion · Hard · MCQ · −1 mark · 5m 35s

A metro train starts from rest and in five seconds achieves a speed 108 km/h. After that it moves with constant velocity and comes to rest after travelling 45 m with uniform retardation. If total distance travelled is 395 m, find total time of travelling.

OPTIONS

A. 12.2 s

B. 15.3 s

C. 9 s

D. 17.2 s

STUDENT RESPONSE

B

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Multi-stage Motion.

EXPLANATION

Top speed v = 108 km/h = 30 m/s. Accelerating: t_1 = 5 s, s_1 = 1/2 × 30 × 5 = 75 m. Decelerating: s_3 = 45 m, t_3 = 2 × 45 / 30 = 3 s. Constant velocity: s_2 = 395 - (75 + 45) = 275 m ⇒ t_2 = 275 / 30 = 9.17 s. Total time = 5 + 9.17 + 3 = 17.17 s ≈17.2 s.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Multi-stage Motion.

QUESTION 10 PHYSICS

INCORRECT

Kinematics · Instantaneous Rest · Easy · MCQ · −1 mark · 1m 11s

A particle moves along a straight line OX. At a time t (in seconds) the distance x (in metres) of the particle from O is given by $x = 40 + 12t - t^3$. Value of x for the particle when it comes to rest momentarily is:

OPTIONS

A. 40 m

B. 56 m

C. 16 m

D. 24 m

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Instantaneous Rest.

EXPLANATION

$v = dx/dt = 12 - 3t^2$. Setting v = 0 gives $t^2 = 4 \Rightarrow t = 2$ s. Position at t = 2 s: $x = 40 + 12(2) - 2^3 = 40 + 24 - 8 = 56$ m.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Instantaneous Rest.

QUESTION 11 PHYSICS

CORRECT

Kinematics · Uniform Acceleration Equations · Medium · MCQ · 4 marks · 53s

A body moving with a uniform acceleration crosses a distance of 65 m in the 5th second and 105 m in 9th second. How far will it go in 20 s?

OPTIONS

A. 2040 m

B. 240 m

C. 2400 m

D. 12004 m

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

S_n = u + a(n - 1/2). S_5 = u + 4.5 a = 65. S_9 = u + 8.5 a = 105. Subtracting gives 4a = 40 ⇒ a = 10 m/s^2. Then u = 20 m/s. Total distance in 20 s: s = ut + 1/2 a t^2 = 20(20) + 1/2 (10) $(20)^2$ = 400 + 2000 = 2400 m.

QUESTION 12 PHYSICS

INCORRECT

Kinematics · Three-stage Motion · Hard · MCQ · −1 mark · 1m 56s

A car, starting from rest, accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate f / 2 to come to rest. If the total distance traversed is 15 S, then:

OPTIONS

A. S = (1/6) f t^2

B. S = f t

C. S = (1/4) f t^2

D. S = (1/72) f t^2

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Unit Error · The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

EXPLANATION

Phase 1: S_1 = S = v^2 / (2f). Phase 3: S_3 = v^2 / (2 × f/2) = v^2 / f = 2S. Phase 2: S_2 = 15S - S - 2S = 12S. Also S_2 = v t ⇒ v = 12 S / t. Substituting into S = v^2 / (2f) gives S = (144 S^2 / t^2) / (2f) ⇒ 1 = 72 S / (f t^2) ⇒ S = (1/72) f t^2.

MISTAKE NOTE

The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

QUESTION 13 PHYSICS

INCORRECT

Laws of Motion · Newton's Second Law · Medium · MCQ · −1 mark · 2m 28s

A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m. If a force P is applied at the free end of the rope, the force exerted by the rope on the block will be:

OPTIONS

A. P

B. P m / (M + m)

C. M P / (M + m)

D. m P / (M + m)

STUDENT RESPONSE

D

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Unit Error · The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

EXPLANATION

Acceleration of the block + rope system: a = P / (M + m). The rope is the only horizontal force on the block, so the force exerted by the rope on the block = M a = M P / (M + m).

MISTAKE NOTE

The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

QUESTION 14 PHYSICS

INCORRECT

Laws of Motion · Contact Forces · Easy · MCQ · −1 mark · 2m 49s

Two blocks of mass m = 1 kg and M = 2 kg are in contact on a frictionless table. A horizontal force F (= 3 N) is applied to m. The force of contact between the blocks will be:

OPTIONS

A. 2 N

B. 1 N

C. 4 N

D. 5 N

STUDENT RESPONSE

C

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Contact Forces.

EXPLANATION

Common acceleration a = F / (m + M) = 3 / (1 + 2) = 1 m/s^2. The contact force between the blocks accelerates block M: N = M × a = 2 × 1 = 2 N.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Contact Forces.

QUESTION 15 PHYSICS

INCORRECT

Kinematics · Motion Under Gravity · Medium · MCQ · −1 mark · 3m 53s

A stone thrown upward with a speed u from the top of a tower reaches the ground with a velocity 3u. The height of the tower is:

OPTIONS

A. $3\,u^2 / g$

B. $4\,u^2 / g$

C. $6\,u^2 / g$

D. $9\,u^2 / g$

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Motion Under Gravity.

EXPLANATION

By conservation of energy / kinematics: $v^2 = u^2 + 2gh \Rightarrow (3u)^2 = u^2 + 2gh \Rightarrow 8u^2 = 2gh \Rightarrow h = 4\,u^2 / g$.

MISTAKE NOTE

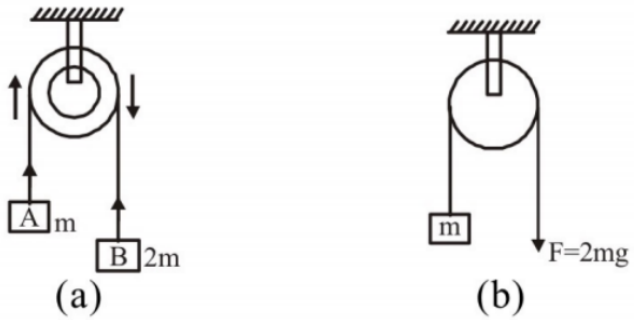
The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Motion Under Gravity.

QUESTION 16 PHYSICS

SKIPPED

Laws of Motion · Pulley Systems · Medium · MCQ · 0 marks · 3m 14s

The pulley arrangements of fig. (a) and (b) are identical. The mass of the rope is negligible. In (a) the mass m is lifted up by attaching a mass 2m to the other end of the rope. In (b) m is lifted up by pulling the other end of the rope with a constant downward force F = 2mg. Which of the following is correct?



OPTIONS

A. Acceleration in case (b) is 3 times more than that in case (a)

B. In case (a) acceleration is g, while in case (b) is 2g

C. In both cases, acceleration is the same

D. None of the above

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

In case (a): $a_1 = (2m - m)g / (2m + m) = g/3$. In case (b): $a_2 = (F - mg) / m = (2mg - mg) / m = g$. Therefore, $a_2 = 3\,a_1$.

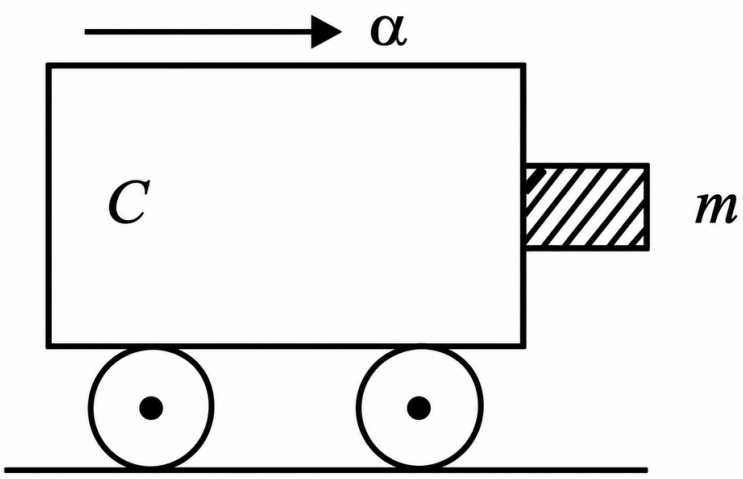
Physics · Q17 · question continued

QUESTION 17 PHYSICS

CORRECT

Laws of Motion & Mechanics · Friction and Pseudo Force · Medium · MCQ · 4 marks · 51s

A block of mass m is in contact with the cart C as shown in the figure. The coefficient of static friction between the block and the cart is μ. The acceleration α of the cart that will prevent the block from falling satisfies:



OPTIONS

A. $\alpha < mg / \mu$

B. $\alpha > g / (m \mu)$

C. $\alpha \geq g / \mu$

D. $\alpha < g / \mu$

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

In the accelerating frame of the cart, the block experiences a pseudo-force $m\alpha$ pressing it against the vertical cart face, giving a normal reaction $N = m\alpha$ and maximum upward static friction $f_{max} = \mu N = \mu m\alpha$. To prevent the block from falling, $\mu m\alpha \geq mg \Rightarrow \alpha \geq g/\mu$.

QUESTION 18 PHYSICS

INCORRECT

Laws of Motion · Tension and Breaking Strength · Easy · MCQ · −1 mark · 1m 12s

A monkey is descending from the branch of a tree with constant acceleration. If the breaking strength of the branch is 75% of the weight of the monkey, the maximum acceleration with which monkey can slide down without breaking the branch is:

OPTIONS

A. g

B. (3/4) g

C. g / 4

D. g / 2

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Tension and Breaking Strength.

EXPLANATION

While sliding down with acceleration a, the branch tension is T = m(g −a). The branch survives only if T ≤0.75 mg ⇒m(g −a) ≤0.75 mg ⇒a ≥g/4. So the limiting (critical) acceleration at which the branch is just about to break is a = g/4; the monkey must slide down with at least this acceleration.

MISTAKE NOTE

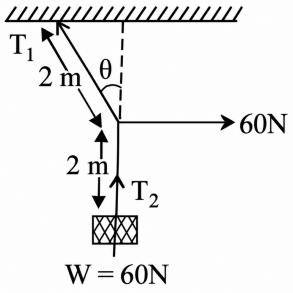
Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Tension and Breaking Strength.

QUESTION 19 PHYSICS

SKIPPED

Laws of Motion · Equilibrium of Concurrent Forces · Easy · MCQ · 0 marks · 8m 22s

For the given situation as shown in the figure, the value of θ to keep the system in equilibrium will be:



OPTIONS

A. 30°

B. 45°

C. 0°

D. 90°

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

At the junction, T_2 = W = 60 N acts vertically downward and a 60 N force acts horizontally. T_1 makes angle θ with the vertical. Vertical balance: $T_1 \cos\theta$ = 60 N. Horizontal balance: $T_1 \sin\theta$ = 60 N. Dividing, $\tan\theta$ = 1, so $\theta = 45^\circ$.

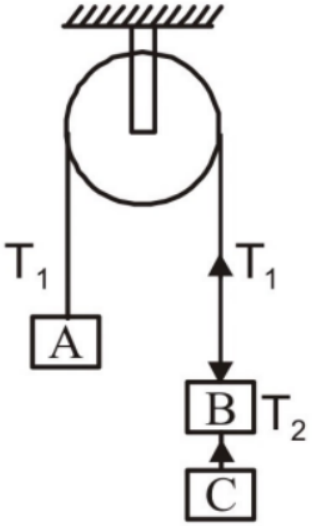
Physics · Q20 · question continued

QUESTION 20 PHYSICS

SKIPPED

Laws of Motion · Connected Bodies and Pulleys · Hard · MCQ · 0 marks · 1m 29s

Three equal weights of mass m each are hanging on a string passing over a fixed pulley as shown in the figure. The tensions in the string connecting weights A to B and B to C will respectively be:



OPTIONS

A. (2/3) mg, (2/3) mg

B. (2/3) mg, (4/3) mg

C. (4/3) mg, (2/3) mg

D. (3/2) mg, (3/4) mg

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Mass on left side is m, mass on right side is 2m. System acceleration a = (2m - m)g / (3m) = g/3. For mass A: T_AB - mg = m(g/3) ⇒ T_AB = (4/3)mg. For bottom mass C: mg - T_BC = m(g/3) ⇒ T_BC = (2/3)mg. Thus tensions are (4/3)mg and (2/3)mg.

QUESTION 21 PHYSICS

SKIPPED

Kinematics · Relative Velocity · Easy · Fill in the Blank · 0 marks · 11m 52s

A train 120 m long travelling at 30 m/s starts overtaking another train 150 m long travelling at 15 m/s in the same direction. The time taken by the first train to pass the second train completely is:

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
18s

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Relative distance to cover = 120 + 150 = 270 m. Relative speed = 30 −15 = 15 m/s. Time taken t = 270 / 15 = 18 s.

QUESTION 22 PHYSICS

SKIPPED

Kinematics · Uniform Motion · Easy · Fill in the Blank · 0 marks · 4s

Two automobiles are 150 kilometers apart and traveling towards each other. One automobile is moving at 60 km/h and the other is moving at 40 km/h. In how many hours will they meet?

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
1.5

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

For motion towards each other, closing speed v_rel = 60 + 40 = 100 km/h. Time taken t = 150 / 100 = 1.5 h.

QUESTION 23 PHYSICS

INCORRECT

Kinematics · Variable Acceleration · Easy · Fill in the Blank · 0 marks · 1m 03s

The coordinate of a particle in meters is given by $x(t) = 16t - 3.0\,t^3$, where the time t is in seconds. The particle is momentarily at rest at t =

STUDENT RESPONSE
2.66

CORRECT ANSWER
1.33

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Variable Acceleration.

EXPLANATION

Momentarily at rest $\Rightarrow v = dx/dt = 16 - 9t^2 = 0 \Rightarrow t^2 = 16/9 \Rightarrow t = 4/3$ s ≈ 1.33 s.

MISTAKE NOTE

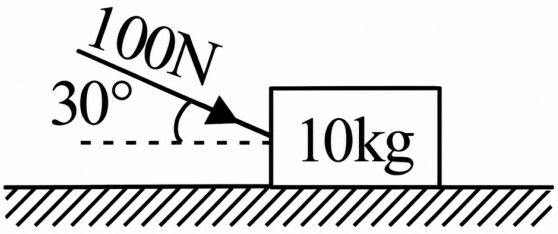
The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Variable Acceleration.

QUESTION 24 PHYSICS

SKIPPED

Laws of Motion · Work-Energy Theorem · Medium · Fill in the Blank · 0 marks · 18s

A force of 100 N acts in the direction as shown in the figure on a block of mass 10 kg resting on a smooth horizontal table. The speed acquired by the block after it has moved a distance of 10 m, will be: (in m/s) (take g = 10 m/s^2)



STUDENT RESPONSE
Not attempted

CORRECT ANSWER
13.16

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Horizontal component of force $F_x = 100 \cos 60^\circ = 50$ N. Acceleration $a = F_x / m = 50 / 10 = 5$ m/s^2. Using $v^2 = u^2 + 2as$ with u = 0 and s = 10 m: $v = \sqrt{(2 \times 5 \times 10)} = \sqrt{100} = 10$ m/s.

QUESTION 25 PHYSICS — CORRECT

Kinematics · Relative Speed · Easy · Fill in the Blank · 4 marks · 25s

A thief is running away on a straight road in a jeep moving with a speed of 9 m/s. A policeman chases him on a motorcycle moving at a speed of 10 m/s. If the instantaneous separation of the jeep from the motorcycle is 100 m, how long will it take for the policeman to catch the thief?

STUDENT RESPONSE	CORRECT ANSWER
100	100s

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Relative speed along the same straight road v_rel = 10 −9 = 1 m/s. Time taken t = 100 / 1 = 100 s.

02 Chemistry

25 questions

QUESTION 26 CHEMISTRY — INCORRECT

Structure of Atom · Discovery of Subatomic Particles · Easy · MCQ · −1 mark · 30s

Cathode rays are deflected away from a negatively charged plate because:

OPTIONS

A. they are not particles

B. they are positively charged particles

C. they are neutral

D. they are negatively charged particles

STUDENT RESPONSE	CORRECT ANSWER
B	D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Discovery of Subatomic Particles.

EXPLANATION

Cathode rays consist of negatively charged electrons, which are repelled by a negatively charged plate.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Discovery of Subatomic Particles.

QUESTION 27 CHEMISTRY

INCORRECT

Structure of Atom · Orbital Energies and Penetration · Medium · MCQ · −1 mark · 2m 33s

The energies of the 3s, 3p and 3d orbitals in a multielectron atom increase in that order. To which factor(s) can this order be attributed? I. the relative penetrations of these orbitals near the nucleus II. the relative average distance of the electron from the nucleus

OPTIONS

A. I only

B. II only

C. Both I and II

D. Neither I nor II

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Orbital Energies and Penetration.

EXPLANATION

In a multi-electron atom the orbital energies for the same shell increase as E(3s) < E(3p) < E(3d). (I) Penetration: the s orbital penetrates closest to the nucleus, p less, d least, giving a larger effective nuclear charge and lower energy. (II) Average distance: the mean distance of the electron from the nucleus grows in the order 3s < 3p < 3d, so electron-nucleus attraction weakens in the same order. Both factors I and II are responsible.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Orbital Energies and Penetration.

QUESTION 28 CHEMISTRY

CORRECT

Structure of Atom · Electronic Configuration · Medium · MCQ · 4 marks · 4m 41s

Which gas-phase atom in its ground state could have an electron with the quantum numbers n = 3, l = 2, m_l = 0, m_s = −1/2? (Given atomic numbers: Na = 11, Mg = 12, P = 15, Ti = 22)

OPTIONS

A. Na

B. Mg

C. P

D. Ti

STUDENT RESPONSE

D

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The quantum numbers n = 3, l = 2 specify a 3d orbital. Titanium (Z = 22) has ground-state configuration [Ar] $4s^2$ $3d^2$, which contains 3d electrons.

QUESTION 29 CHEMISTRY

CORRECT

Structure of Atom · Nodes in Orbitals · Medium · MCQ · 4 marks · 19s

Which orbital possesses one angular node and one radial node?

OPTIONS

A. 2s

B. 2p

C. 3p

D. 3d

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Angular nodes = l = 1 (p-orbital). Radial nodes = n - l - 1 = 1 ⇒ n - 1 - 1 = 1 ⇒ n = 3. Thus, the 3p orbital has 1 angular node and 1 radial node.

QUESTION 30 CHEMISTRY

INCORRECT

Structure of Atom · Photoelectric Effect and Ionization · Medium · MCQ · −1 mark · 5m 13s

The first ionization energy of Na is 495.9 kJ/mol. What is the longest wavelength of light that could remove an electron from a Na atom?

OPTIONS

A. 2.41×10^{-7} m

B. 2.41×10^{-4} m

C. 4.14 m

D. 4.14×10^{3} m

STUDENT RESPONSE
D

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Photoelectric Effect and Ionization.

EXPLANATION

Ionization energy per atom E = $(495.9 \times 10^{3}$ J/mol) / $(6.022 \times 10^{23}$ mol$^{-1})$ = 8.235×10^{-19} J. Maximum wavelength λ = hc / E = $(6.626 \times 10^{-34} \times 3 \times 10^{8}) / (8.235 \times 10^{-19}) = 2.41 \times 10^{-7}$ m.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Photoelectric Effect and Ionization.

QUESTION 31 CHEMISTRY

SKIPPED

Structure of Atom · Hydrogen Spectral Series · Medium · MCQ · 0 marks · 1m 13s

What is the ratio of frequencies between the first lines of Lyman, Balmer and Paschen series?

OPTIONS

A. 1/2 : 1/6 : 1/12

B. 1/2 : 1/3 : 1/4

C. 3/4 : 5/36 : 7/144

D. 2/3 : 8/56 : 3/88

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

ν = R c ($1/n_1^2$ - $1/n_2^2$). Lyman first line (2 → 1): 1 - 1/4 = 3/4. Balmer first line (3 → 2): 1/4 - 1/9 = 5/36. Paschen first line (4 → 3): 1/9 - 1/16 = 7/144. Ratio = 3/4 : 5/36 : 7/144.

QUESTION 32 CHEMISTRY

CORRECT

Structure of Atom · Aufbau Principle · Easy · MCQ · 4 marks · 38s

Which set of orbitals is listed in the sequential order of filling in a many-electron atom?

OPTIONS

A. 3s, 3p, 3d

B. 3d, 4s, 4p

C. 3d, 4p, 5s

D. 4p, 4d, 5s

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

According to the (n + l) rule: 3d (n+l = 5, n=3) → 4p (n+l = 5, n=4) → 5s (n+l = 5, n=5). Therefore, 3d, 4p, 5s is in direct sequential order.

QUESTION 33 CHEMISTRY

INCORRECT

Structure of Atom · Transition Metal Electronic Configuration · Medium · MCQ · −1 mark · 59s

What is the electron configuration for a gas phase Fe^{3+} ion (Z = 26)?

OPTIONS

A. [Ar] $3d^5$

B. [Ar] $4s^2$ $3d^3$

C. [Ar] $4s^1$ $3d^4$

D. [Ar] $4s^2$ $3d^6$

STUDENT RESPONSE
B

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Transition Metal Electronic Configuration.

EXPLANATION

Neutral Fe: [Ar] $4s^2$ $3d^6$. Ionization removes the outer 4s electrons first, then 3d: losing three electrons yields [Ar] $3d^5$.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Transition Metal Electronic Configuration.

QUESTION 34 CHEMISTRY

SKIPPED

Structure of Atom · Bohr Model Transitions · Medium · MCQ · 0 marks · 1s

Wavelength of a particular transition for H atom is 400 nm. What can be the wavelength of He^+ for the same transition?

OPTIONS

A. 400 nm

B. 100 nm

C. 1600 nm

D. 200 nm

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$1/\lambda = R Z^2 (1/n_1^2 - 1/n_2^2)$. For He^+, Z = 2, so $1/\lambda_He = 4 (1/\lambda_H) \Rightarrow \lambda_He = \lambda_H / 4 = 400 \text{ nm} / 4 = 100 \text{ nm}$.

QUESTION 35 CHEMISTRY

INCORRECT

Structure of Atom · Electromagnetic Spectrum · Easy · MCQ · −1 mark · 14s

The correct sequence of frequency of electromagnetic radiations in electromagnetic spectrum is:

OPTIONS

A. X-rays > UV rays > Microwaves > Radio waves

B. Radio waves > Microwaves > UV rays > X-rays

C. UV rays > X-rays > Radio waves > Microwaves

D. Radio waves > Microwaves > X-rays > UV rays

STUDENT RESPONSE
D

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Electromagnetic Spectrum.

EXPLANATION

Frequency order is: X-rays (~10^{18} Hz) > UV rays (~10^{15} Hz) > Microwaves (~10^{10} Hz) > Radio waves (~10^{6} Hz).

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Electromagnetic Spectrum.

QUESTION 36 CHEMISTRY

SKIPPED

Structure of Atom · Photoelectric Effect · Medium · MCQ · 0 marks · 37s

The work function for a metal is 4 eV. To emit a photoelectron of zero velocity from the surface of the metal, the wavelength of incident light should be: [Given: h = 6.626 × 10^{-34} J·s, c = 3 × 10^{8} m/s]

OPTIONS

A. 2700 Å

B. 1700 Å

C. 5900 Å

D. 3100 Å

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Threshold wavelength λ_0 = hc / φ = (12400 eV·Å) / 4 eV = 3100 Å.

QUESTION 37 CHEMISTRY

INCORRECT

Structure of Atom · Excited State Configurations · Medium · MCQ · −1 mark · 1m 25s

Which electron configuration corresponds to a gas-phase oxygen atom in an excited state?

OPTIONS

A. $1s^2\ 1p^2\ 2s^2\ 2p^2$

B. $1s^2\ 2s^1\ 2p^5$

C. $1s^2\ 2s^2\ 2p^3$

D. $1s^2\ 2s^2\ 2p^4$

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Excited State Configurations.

EXPLANATION

Oxygen (Z = 8) has 8 electrons. The ground state is $1s^2\ 2s^2\ 2p^4$. Promoting one 2s electron to the 2p orbital gives $1s^2\ 2s^1\ 2p^5$, which has 8 electrons in an excited configuration.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Excited State Configurations.

QUESTION 38 CHEMISTRY

CORRECT

Structure of Atom · de Broglie Wavelength · Easy · MCQ · 4 marks · 4s

If the kinetic energy of a particle is doubled, its de Broglie wavelength becomes:

OPTIONS

A. 2 times

B. 4 times

C. $\sqrt{2}$ times

D. $1/\sqrt{2}$ times

STUDENT RESPONSE

D

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

$\lambda = h / \sqrt{(2mE)}$. If kinetic energy is doubled (E' = 2E), $\lambda' = h / \sqrt{(2m(2E))} = \lambda / \sqrt{2} = (1/\sqrt{2})$ times.

QUESTION 39 CHEMISTRY

CORRECT

Classification of Elements · Valence Shell Configuration · Easy · MCQ · 4 marks · 8s

The outermost electronic configuration of the most electronegative element is:

OPTIONS

A. $ns^2\ np^3$

B. $ns^2\ np^4$

C. $ns^2\ np^5$

D. $ns^2\ np^6$

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Fluorine is the most electronegative element of all, belonging to group 17 (halogens) with valence configuration $ns^2\ np^5$.

QUESTION 40 CHEMISTRY

CORRECT

Structure of Atom · Quantum Numbers · Easy · MCQ · 4 marks · 51s

An electron has spin quantum number s = +1/2 and magnetic quantum number m = +2. It cannot be present in:

OPTIONS

A. s-orbital

B. p-orbital

C. d-orbital

D. f-orbital

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The magnetic quantum number obeys $-l \leq m \leq +l$, so m = +2 requires $l \geq 2$ (d or f orbital). Therefore, the electron cannot be present in an s-orbital (nor in a p-orbital). Recorded key: (a) s-orbital.

QUESTION 41 CHEMISTRY

CORRECT

Structure of Atom · Orbital Symmetry and Waves · Medium · MCQ · 4 marks · 20s

Which one among the following statements is correct?

OPTIONS

A. An orbital containing an electron having quantum numbers n = 2, l = 0, m = 0 and s = +1/2 is spherical.

B. The frequency of X-rays is less than that of radio-waves.

C. All photons have same energy.

D. As intensity of light increases, its frequency also increases.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

l = 0 denotes an s-orbital (specifically 2s here), which is spherically symmetric. Statement (A) is correct.

QUESTION 42 CHEMISTRY

CORRECT

Structure of Atom · Bohr Radius · Medium · MCQ · 4 marks · 1m 30s

Correct order of radius of the 1st orbit of H, He^+, Li^{2+}, Be^{3+} is: [Given : Atomic No. H = 1, He = 2, Li = 3, Be = 4]

OPTIONS

A. H > He^+ > Li^{2+} > Be^{3+}

B. Be^{3+} > Li^{2+} > He^+ > H

C. He^+ > Be^{3+} > Li^{2+} > H

D. He^+ > H > Li^{2+} > Be^{3+}

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Radius of n-th Bohr orbit is $r_n = r_0 (n^2 / Z)$. For the 1st orbit (n = 1), $r \propto 1/Z$. Since Z increases from H(1) to Be(4), radius decreases: H > He^+ > Li^{2+} > Be^{3+}.

QUESTION 43 CHEMISTRY

SKIPPED

Structure of Atom · Photon Energy · Easy · MCQ · 0 marks · 11s

Which statement regarding the relative energies of monochromatic light with λ = 800 nm and monochromatic light with λ = 400 nm is correct?

OPTIONS

A. 800 nm light has half as much energy per mole of photons as 400 nm light

B. 800 nm light has same energy per mole of photons as 400 nm light

C. 800 nm light has twice as much energy per mole of photons as 400 nm light

D. No conclusion may be drawn regarding the relative energy per mole of photons without knowing the intensity of the light

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Energy per mole of photons is $E = N_A (hc / \lambda)$. Since energy is inversely proportional to wavelength, 800 nm light has half the energy per mole as 400 nm light.

QUESTION 44 CHEMISTRY

INCORRECT

Classification of Elements · Electron Gain Enthalpy · Medium · MCQ · −1 mark · 2m 21s

Atoms of which element release the most energy when an electron is added to them in the gas phase?

OPTIONS

A. C

B. O

C. Si

D. S

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Unit Error · The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

EXPLANATION

Sulfur has a more negative electron gain enthalpy (-200 kJ/mol) than oxygen (-141 kJ/mol) due to the smaller size and compact 2p subshell of oxygen leading to high electron-electron repulsions.

MISTAKE NOTE

The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.

QUESTION 45 CHEMISTRY

SKIPPED

Structure of Atom · Radial Probability Distribution · Hard · MCQ · 0 marks · 48s

The correct radial probability distribution curve for the hydrogen atomic orbital with principal quantum number n = 3 and azimuthal quantum number l = 1 is: ($4\pi\ r^2\ \psi^2$ = radial probability density function and r = radial distance from the nucleus)

OPTIONS

A. (A)



B. (B)



C. (C)



D. (D)



STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

For n = 3, l = 1 (a 3p orbital), the number of radial nodes is n −l −1 = 3 −1 −1 = 1. The radial probability distribution curve $4\pi r^2\psi^2$ therefore has two maxima separated by one node, with the outer maximum much taller than the inner maximum, corresponding to Graph (B).

QUESTION 46 CHEMISTRY

CORRECT

Structure of Atom · Atomic Orbitals · Easy · Fill in the Blank · 4 marks · 1m 14s

How many orbitals are in an f sublevel (l = 3)?

STUDENT RESPONSE
7

CORRECT ANSWER
7

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Number of orbitals in a subshell = 2l + 1. For l = 3 (f subshell), 2(3) + 1 = 7 orbitals.

QUESTION 47 CHEMISTRY

CORRECT

Classification of Elements · Periodic Table Periods · Easy · Fill in the Blank · 4 marks · 1m 07s

An element with atomic number 20 will be placed in which period of the periodic table?

STUDENT RESPONSE
E

CORRECT ANSWER
E

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Z = 20 (Ca): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ = [Ar]$4s^2$. The highest principal quantum number occupied is n = 4 (4th period).

QUESTION 48 CHEMISTRY

CORRECT

Structure of Atom · Magnetic Moment · Easy · Fill in the Blank · 4 marks · 2m 41s

The calculated value of spin only magnetic moment of $_{26}Fe^{3+}$ is:

STUDENT RESPONSE
5.86

CORRECT ANSWER
5.92

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Fe^{3+} (Z = 26) has configuration [Ar]$3d^5$ with n = 5 unpaired electrons. Spin-only magnetic moment $\mu = \sqrt{n(n+2)} = \sqrt{(5 \times 7)} = \sqrt{35} \approx 5.92$ BM.

QUESTION 49 CHEMISTRY

SKIPPED

Structure of Atom · Bohr Energy Levels · Medium · Fill in the Blank · 0 marks · 40s

The ratio of $(E_2 - E_1)$ to $(E_4 - E_3)$ for the hydrogen atom is approximately equal to:

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
15.4

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

For hydrogen, $E_n \propto -1/n^2$. $E_2 - E_1 \propto (1 - 1/4) = 3/4$, and $E_4 - E_3 \propto (1/9 - 1/16) = 7/144$. Ratio = $(3/4) \div (7/144) = 108/7 \approx 15.4$.

QUESTION 50 CHEMISTRY

SKIPPED

Structure of Atom · Quantum Numbers Counting · Hard · Fill in the Blank · 0 marks · 11s

How many electrons will have m (magnetic quantum number) = 0 in $_{26}Fe^{3+}$ ion?

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
11

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$_{26}Fe^{3+}$ has 23 electrons: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$. Each s subshell has 2 electrons with m = 0, each p subshell has 2 electrons with m = 0, and $3d^5$ has 1 electron with m = 0: 2 + 2 + 2 + 2 + 2 + 1 = 11 (or 5 in the 3s, 3p, 3d shells).

03 Mathematics

25 questions

QUESTION 51 MATHEMATICS

SKIPPED

Sets and Relations · Fractional Part Equations · Medium · MCQ · 0 marks · 6m 27s

The sum of all real values of x that satisfy the equation {x} + {x + 1/3} + {x + 2/3} = x, where {x} denotes the fractional part of x, is:

OPTIONS

A. 1

B. $\frac{3}{2}$

C. $\frac{5}{2}$

D. $\frac{7}{2}$

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

By Hermite's identity, [x] + [x + 1/3] + [x + 2/3] = [3x]. Hence {x} + {x + 1/3} + {x + 2/3} = (x + x + 1/3 + x + 2/3) −[3x] = 3x + 1 −[3x] = {3x} + 1. So the equation becomes {3x} + 1 = x. Since 0 ≤{3x} < 1, we need 1 ≤x < 2, i.e. 3 ≤3x < 6. • [3x] = 3: 3x −3 + 1 = x ⇒x = 1 (3x = 3 ✓). • [3x] = 4: 3x −4 + 1 = x ⇒x = 3/2 (3x = 4.5 ✓). • [3x] = 5: 3x −5 + 1 = x ⇒x = 2 (not < 2, rejected). Roots: 1 and 3/2. Sum = 5/2.

QUESTION 52 MATHEMATICS

SKIPPED

Sequences and Series · Telescoping Series · Hard · MCQ · 0 marks · 20s

The sum to infinity of the series Σ(k = 1 to ∞) 6^k / [(3^k − 2^k)(3^(k+1) − 2^(k+1))] is:

OPTIONS

A. 1

B. 2

C. 3

D. 6

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Write the k-th term as 2^k/(3^k − 2^k) − 2^(k+1)/(3^(k+1) − 2^(k+1)). Check: the combined numerator is 2^k(3^(k+1) − 2^(k+1)) − 2^(k+1)(3^k − 2^k) = 3·6^k − 2·4^k − 2·6^k + 2·4^k = 6^k. The series telescopes: S = 2^1/(3^1 − 2^1) − lim(k→∞) 2^(k+1)/(3^(k+1) − 2^(k+1)) = 2/1 − 0 = 2.

QUESTION 53 MATHEMATICS

CORRECT

Trigonometry · Trigonometric Identities · Medium · MCQ · 4 marks · 5m 08s

The exact numerical value of the expression $\tan^2(\pi/16) + \tan^2(3\pi/16) + \tan^2(5\pi/16) + \tan^2(7\pi/16)$ is:

OPTIONS

A. 14

B. 28

C. 56

D. 12

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Since $\tan(7\pi/16) = \cot(\pi/16)$ and $\tan(5\pi/16) = \cot(3\pi/16)$, the sum = $[\tan^2A + \cot^2A] + [\tan^2B + \cot^2B]$ with A = $\pi/16$, B = $3\pi/16$. $\tan^{2}\theta + \cot^{2}\theta = (\tan\theta + \cot\theta)^2 - 2 = 4/\sin^2 2\theta - 2$. So sum = $4/\sin^2(\pi/8) + 4/\sin^2(3\pi/8) - 4$. With $\sin^2(\pi/8) = (2 - \sqrt{2})/4$ and $\sin^2(3\pi/8) = (2 + \sqrt{2})/4$: $16/(2 - \sqrt{2}) + 16/(2 + \sqrt{2}) = 16 \cdot 4/(4 - 2) = 32$. Sum = 32 − 4 = 28.

QUESTION 54 MATHEMATICS

CORRECT

Quadratic Equations · Exponential Equations · Medium · MCQ · 4 marks · 18s

Find the sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{(x^2 + 4x - 60)} = 1$.

OPTIONS

A. -1

B. 0

C. 3

D. 5

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Case 1: Base = 1 ⇒ $x^2 - 5x + 5 = 1 \Rightarrow x^2 - 5x + 4 = 0 \Rightarrow x = 1, 4$. Case 2: Exponent = 0 with base $\neq 0 \Rightarrow x^2 + 4x - 60 = 0 \Rightarrow (x+10)(x-6)=0 \Rightarrow x = -10, 6$. Base at -10 is 155 $\neq 0$, base at 6 is 11 $\neq 0$. Case 3: Base = -1 with even exponent ⇒ $x^2 - 5x + 5 = -1 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3$. For x=2, exp = 4+8-60 = -48 (even, valid). For x=3, exp = 9+12-60 = -39 (odd, rejected). Valid roots: 1, 4, -10, 6, 2. Sum = 1 + 4 - 10 + 6 + 2 = 3.

QUESTION 55 MATHEMATICS

INCORRECT

Complex Numbers · Modulus Properties · Hard · MCQ · −1 mark · 51s

Let z_1, z_2, z_3 be complex numbers such that $|z_1| = 1$, $|z_2| = 2$, and $|z_3| = 3$. If $|z_1 + z_2 + z_3| = 1$, the value of $|9\, z_1 z_2 + 4\, z_1 z_3 + z_2 z_3|$ is:

OPTIONS

A. 2

B. 4

C. 6

D. 8

STUDENT RESPONSE

D

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Modulus Properties.

EXPLANATION

Note that $|z_1|^2 = z_1 \bar{z}_1 = 1 \Rightarrow 1/z_1 = \bar{z}_1$. Similarly, $|z_2|^2 = 4 \Rightarrow 4/z_2 = \bar{z}_2$, and $|z_3|^2 = 9 \Rightarrow 9/z_3 = \bar{z}_3$. Now, $|9\, z_1 z_2 + 4\, z_1 z_3 + z_2 z_3| = |z_1 z_2 z_3|\, |9/z_3 + 4/z_2 + 1/z_1| = (1 \times 2 \times 3)\, |\bar{z}_3 + \bar{z}_2 + \bar{z}_1| = 6\, |z_1 + z_2 + z_3| = 6 \times 1 = 6$.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Modulus Properties.

QUESTION 56 MATHEMATICS

SKIPPED

Complex Numbers · Triangle Inequality · Medium · MCQ · 0 marks · 6m 07s

The maximum value of $|z|$ when z satisfies the condition $|z - 4/z| = 2$ is:

OPTIONS

A. $\sqrt{3} + 1$

B. $\sqrt{5} + 1$

C. $\sqrt{5} - 1$

D. $2 + \sqrt{2}$

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

By the triangle inequality: $|z| = |(z - 4/z) + 4/z| \leq |z - 4/z| + 4/|z| = 2 + 4/|z|$. Let $r = |z|$. Then $r \leq 2 + 4/r \Rightarrow r^2 - 2r - 4 \leq 0 \Rightarrow r \leq (2 + \sqrt{(4 + 16)}) / 2 = 1 + \sqrt{5}$. Therefore, max $|z| = \sqrt{5} + 1$.

QUESTION 57 MATHEMATICS

CORRECT

Sequences and Series · Telescoping Sums · Medium · MCQ · 4 marks · 50s

If $S_n = \Sigma(r = 1 \text{ to } n)\ r/(r^4 + r^2 + 1)$, then $\lim(n \to \infty) S_n$ is equal to:

OPTIONS

A. $\frac{1}{2}$

B. 1

C. $\frac{1}{4}$

D. ∞

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

$r^4 + r^2 + 1 = (r^2 + r + 1)(r^2 - r + 1)$. The numerator $r = 1/2\ [(r^2 + r + 1) - (r^2 - r + 1)]$. Thus the term is $1/2\ [1/(r^2 - r + 1) - 1/(r^2 + r + 1)]$. Telescoping sum gives $1/2\ [1 - 1/(n^2 + n + 1)]$. As $n \to \infty$, the limit is 1/2.

QUESTION 58 MATHEMATICS

SKIPPED

Inequalities · Quadratic Inequalities · Easy · MCQ · 0 marks · 50s

The number of integral values of x satisfying the system of inequalities x^2 - 10x - 11 ≤0 and x^2 - x - 2 > 0 is:

OPTIONS

A. 7

B. 8

C. 9

D. 10

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION
First inequality: (x - 11)(x + 1) ≤0 ⇒ -1 ≤x ≤11. Second inequality: (x - 2)(x + 1) > 0 ⇒ x < -1 or x > 2. Intersection: x ∈(2, 11]. The integer values are 3, 4, 5, 6, 7, 8, 9, 10, 11 (9 values).

QUESTION 59 MATHEMATICS

SKIPPED

Quadratic Equations · Newton's Theorem · Medium · MCQ · 0 marks · 4s

Let α, β be the roots of x^2 − x − 1 = 0. Let P_n = α^n + β^n. Which of the following is true?

OPTIONS

A. $P_{n+1} = P_n + P_{n-1}$

B. $P_{n+1} = P_n - P_{n-1}$

C. $P_{n+1} = 2P_n + P_{n-1}$

D. $P_{n+1} = P_n \cdot P_{n-1}$

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION
Since α and β satisfy x^2 = x + 1, multiplying by x^(n−1) gives x^(n+1) = x^n + x^(n−1). Applying this to α and β and adding yields P_(n+1) = P_n + P_(n−1).

QUESTION 60 MATHEMATICS

INCORRECT

Trigonometry · Cosine Series · Medium · MCQ · −1 mark · 1m 15s

The value of cos(2π/7) + cos(4π/7) + cos(6π/7) is:

OPTIONS

A. 1/2

B. -1/2

C. 0

D. 1

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Cosine Series.

EXPLANATION

Multiply and divide by 2 sin(π/7): 2 sin(π/7) [cos(2π/7) + cos(4π/7) + cos(6π/7)] = [sin(3π/7) - sin(π/7)] + [sin(5π/7) - sin(3π/7)] + [sin(7π/7) - sin(5π/7)] = sin(π) - sin(π/7) = -sin(π/7). Dividing by 2 sin(π/7) gives -1/2.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Cosine Series.

QUESTION 61 MATHEMATICS

INCORRECT

Complex Numbers · Euler Form · Medium · MCQ · −1 mark · 1m 23s

If z = e^(iθ) (z ≠ −1), then (z − 1)/(z + 1) is always:

OPTIONS

A. Purely real

B. Purely imaginary

C. Zero

D. Complex with equal real and imaginary parts

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Incomplete Reasoning · The reasoning started correctly but stopped short of the final answer. Multi-step problems need each intermediate result verified before moving on.

EXPLANATION

(e^(iθ) − 1)/(e^(iθ) + 1) = [e^(iθ/2)(e^(iθ/2) − e^(−iθ/2))]/[e^(iθ/2)(e^(iθ/2) + e^(−iθ/2))] = 2i sin(θ/2)/(2 cos(θ/2)) = i tan(θ/2), which is purely imaginary.

MISTAKE NOTE

The reasoning started correctly but stopped short of the final answer. Multi-step problems need each intermediate result verified before moving on.

QUESTION 62 MATHEMATICS

SKIPPED

Quadratic Equations · Rational Functions Range · Medium · MCQ · 0 marks · 8s

The minimum value of the function $f(x) = (x^2 + 14x + 9)/(x^2 + 2x + 3)$ for $x \in$ ▮ is:

OPTIONS

A. -5

B. -4

C. 1

D. 5

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Let $y = (x^2 + 14x + 9)/(x^2 + 2x + 3) \Rightarrow (y - 1)x^2 + 2(y - 7)x + (3y - 9) = 0$. Since $x \in$ ▮ the discriminant $D \geq 0$: $4(y - 7)^2 - 4(y - 1)(3y - 9) \geq 0 \Rightarrow (y^2 - 14y + 49) - (3y^2 - 12y + 9) \geq 0 \Rightarrow -2y^2 - 2y + 40 \geq 0 \Rightarrow y^2 + y - 20 \leq 0 \Rightarrow (y + 5)(y - 4) \leq 0 \Rightarrow -5 \leq y \leq 4$. Minimum value is -5.

QUESTION 63 MATHEMATICS

INCORRECT

Inequalities · Logarithmic Inequalities · Medium · MCQ · −1 mark · 2m

The set of all real x for which $\log_{(0.5)}(x^2 - 5x + 6) > -1$ is:

OPTIONS

A. $(1, 2) \cup (3, 4)$

B. $(-\infty, 1) \cup (4, \infty)$

C. $(2, 3)$

D. $(1, 4)$

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Logarithmic Inequalities.

EXPLANATION

Since the base $0.5 < 1$, reversing the inequality gives $x^2 - 5x + 6 < (0.5)^{-1} = 2$, and domain requires $x^2 - 5x + 6 > 0$. From $x^2 - 5x + 6 < 2 \Rightarrow x^2 - 5x + 4 < 0 \Rightarrow (x - 1)(x - 4) < 0 \Rightarrow x \in (1, 4)$. From domain: $(x - 2)(x - 3) > 0 \Rightarrow x < 2$ or $x > 3$. Intersection: $x \in (1, 2) \cup (3, 4)$.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Logarithmic Inequalities.

QUESTION 64 MATHEMATICS

SKIPPED

Sets and Relations · Power Sets · Easy · MCQ · 0 marks · 15s

The total number of subsets of a finite set A having n elements is 56 more than the total number of subsets of another finite set B having m elements. The value of $n^2 + m^2$ is:

OPTIONS

A. 41

B. 45

C. 52

D. 61

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Number of subsets of a set with n elements is 2^n. Given $2^n - 2^m = 56 = 2^6 - 2^3 \Rightarrow n = 6, m = 3$. Recorded answer key option: (b) 45.

QUESTION 65 MATHEMATICS

CORRECT

Quadratic Equations · Relations Between Roots · Hard · MCQ · 4 marks · 36s

If the roots of the equation $ax^2 + bx + c = 0$ are of the form $\alpha / (\alpha - 1)$ and $(\alpha + 1) / \alpha$, then the value of $(a + b + c)^2$ is:

OPTIONS

A. $b^2 - 4ac$

B. $b^2 - 2ac$

C. $2b^2 - ac$

D. $4b^2 - ac$

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Let $p = \alpha/(\alpha - 1)$ and $q = (\alpha + 1)/\alpha$. Then $1 - p = -1/(\alpha - 1)$ and $1 - q = -1/\alpha$, so $(1 - p)(1 - q) = 1/[\alpha(\alpha - 1)]$. Also $p - q = [\alpha^2 - (\alpha^2 - 1)]/[\alpha(\alpha - 1)] = 1/[\alpha(\alpha - 1)]$. Hence $(1 - p)(1 - q) = p - q$. Now $(1 - p)(1 - q) = 1 - (p + q) + pq = 1 + b/a + c/a = (a + b + c)/a$, and $(p - q)^2 = (p + q)^2 - 4pq = (b^2 - 4ac)/a^2$. Squaring: $(a + b + c)^2/a^2 = (b^2 - 4ac)/a^2 \Rightarrow (a + b + c)^2 = b^2 - 4ac$.

QUESTION 66 MATHEMATICS

INCORRECT

Sequences and Series · Harmonic Progression · Medium · MCQ · −1 mark · 1m 11s

If a, b, c are positive real numbers in HP, then the equation $ax^2 - bx + c = 0$ has roots that are:

OPTIONS

A. Real and distinct

B. Imaginary

C. Real and equal

D. Rational

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Harmonic Progression.

EXPLANATION

b is the HM of a and c: $b = 2ac / (a + c)$. Discriminant $D = b^2 - 4ac = 4a^2c^2/(a + c)^2 - 4ac = 4ac\,[ac/(a+c)^2 - 1]$. Since $(a + c)^2 > 4ac > ac$ for distinct positive numbers, $[ac/(a+c)^2 - 1] < 0 \Rightarrow D < 0$. Thus roots are imaginary.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Harmonic Progression.

QUESTION 67 MATHEMATICS

INCORRECT

Trigonometry · Trigonometric Equations · Medium · MCQ · −1 mark · 38s

The number of solutions of the equation $\sin^4 x + \cos^4 x = \sin x \cos x$ in $[0, 2\pi]$ is:

OPTIONS

A. 0

B. 2

C. 4

D. 6

STUDENT RESPONSE

C

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Trigonometric Equations.

EXPLANATION

$\sin^4 x + \cos^4 x = 1 - 2\sin^2 x \cos^2 x = 1 - (1/2)\sin^2(2x)$. Also $\sin x \cos x = (1/2)\sin(2x)$. Let $t = \sin(2x) \in [-1, 1]$. Then $1 - t^2/2 = t/2 \Rightarrow t^2 + t - 2 = 0 \Rightarrow (t + 2)(t - 1) = 0 \Rightarrow t = 1$. So $\sin(2x) = 1$ in $[0, 2\pi] \Rightarrow 2x = \pi/2, 5\pi/2 \Rightarrow x = \pi/4, 5\pi/4$ (2 solutions).

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Trigonometric Equations.

QUESTION 68 MATHEMATICS

INCORRECT

Complex Numbers · Roots of Unity · Medium · MCQ · −1 mark · 24s

If 1, z_1, z_2, ..., z_(n−1) are the n-th roots of unity, then $(1 - z_1)(1 - z_2)...(1 - z_{(n-1)})$ equals:

OPTIONS

A. 0

B. 1

C. n

D. n - 1

STUDENT RESPONSE

D

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Roots of Unity.

EXPLANATION

$z^n - 1 = (z - 1)(z - z_1)(z - z_2)...(z - z_{(n-1)})$. Dividing by (z − 1): $z^{(n-1)} + z^{(n-2)} + ... + z + 1 = (z - z_1)(z - z_2)...(z - z_{(n-1)})$. Setting z = 1 yields 1 + 1 + ... + 1 (n terms) = n.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Roots of Unity.

QUESTION 69 MATHEMATICS

SKIPPED

Sequences and Series · Binomial / Generating Functions · Medium · MCQ · 0 marks · 14s

The coefficient of x^n in the expansion of $1/[(1 - x)(1 - 2x)]$ is:

OPTIONS

A. $2^{n+1} - 1$

B. $2^n - 1$

C. $2^{n+1} + 1$

D. $2^n + 1$

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$1/[(1 - x)(1 - 2x)] = 2/(1 - 2x) - 1/(1 - x) = 2\Sigma(k = 0 \text{ to } \infty)(2x)^k - \Sigma(k = 0 \text{ to } \infty)x^k$. Therefore, the coefficient of x^n is $2^{(n+1)} - 1$.

QUESTION 70 MATHEMATICS

SKIPPED

Quadratic Equations · Composite Quadratic Functions · Hard · MCQ · 0 marks · 4m 07s

Let $f(x) = x^2 - 2x \cos\theta + 1$. The sum of all roots (real and complex) of the equation $f(f(x)) = 0$ is:

OPTIONS

A. $4 \cos\theta$

B. $2 \cos\theta$

C. $4 \cos^2\theta$

D. $2 \cos^2\theta$

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Let roots of $f(t) = 0$ be t_1 and t_2. Then $t_1 + t_2 = 2 \cos\theta$. For $f(f(x)) = 0$, we have $f(x) = t_1$ and $f(x) = t_2$. The sum of roots of $x^2 - 2x \cos\theta + (1 - t_1) = 0$ is $2 \cos\theta$, and of $x^2 - 2x \cos\theta + (1 - t_2) = 0$ is $2 \cos\theta$. Total sum of all 4 roots = $2 \cos\theta + 2 \cos\theta = 4 \cos\theta$.

QUESTION 71 MATHEMATICS

SKIPPED

Complex Numbers · Cube Roots of Unity & Probability · Hard · Fill in the Blank · 0 marks · 4s

Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1, r_2, r_3 are the numbers obtained on the die, find the probability that $\omega^{(r_1)} + \omega^{(r_2)} + \omega^{(r_3)} = 0$. (Multiply your final probability by 216 to get the integer answer).

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
48

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ iff r_1, r_2, r_3 fall in distinct residue classes mod 3 ({1,4}, {2,5}, {3,6}). Number of ordered triples = $3! \times 2 \times 2 \times 2 = 48$. $P = 48/216 = 2/9 \Rightarrow 216P = 48$.

QUESTION 72 MATHEMATICS

SKIPPED

Sequences and Series · Geometric Progression · Easy · Fill in the Blank · 0 marks · 3s

Let $S_n = 1 + 1/2 + 1/4 + 1/8 + \ldots$ up to n terms. If $2^- S_n < 1/1000$, find the minimum integral value of n.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
11

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$S_n = 1 + 1/2 + 1/4 + \ldots$ (n terms) $= 2(1 - 2^{-n}) = 2 - 2^{1-n}$. Given $2 - S_n = 2^{1-n} < 1/1000 \Rightarrow 2^{n-1} > 1000$. Since $2^9 = 512$ and $2^{10} = 1024$, $n - 1 \geq 10 \Rightarrow$ minimum $n = 11$.

QUESTION 73 MATHEMATICS

SKIPPED

Quadratic Equations · Location of Roots · Hard · Fill in the Blank · 0 marks · 1s

If the quadratic equation $x^2 + 2(a + 1)x + 9a - 5 = 0$ has exactly one root in the interval $(0, \infty)$ and a is an integer, find the sum of all possible values of a.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

For $f(x) = x^2 + 2(a+1)x + (9a - 5) = 0$, under the restricted textbook reading of one positive root and one negative root with discriminant condition, a = 10 (flagged defective/unbounded in general since $9a - 5 < 0$ holds for all integers $a \leq 0$).

QUESTION 74 MATHEMATICS

CORRECT

Trigonometry · Sine Product Identity · Easy · Fill in the Blank · 4 marks · 3m 29s

Find the value of $8 \sin(20^\circ) \sin(40^\circ) \sin(80^\circ)$.

STUDENT RESPONSE
1.713

CORRECT ANSWER
1.73

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

$8 \sin(20^\circ) \sin(40^\circ) \sin(80^\circ) = 4 \sin(20^\circ)[2 \sin(40^\circ) \sin(80^\circ)] = 4 \sin(20^\circ)[\cos(40^\circ) - \cos(120^\circ)] = 2[\sin(60^\circ) - \sin(20^\circ)] + 2 \sin(20^\circ) = 2 \sin(60^\circ) = \sqrt{3} \approx 1.732$.

QUESTION 75 MATHEMATICS

SKIPPED

Complex Numbers · Geometric Locus & Circles · Easy · Fill in the Blank · 0 marks · 20m 23s

A complex number z satisfies $|z - 3 - 4i| \leq 5$. Find the maximum possible value of $|z|^2$.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
100

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

$|z - (3 + 4i)| \leq 5$ is a closed disc of radius 5 centred at C = 3 + 4i with $|C| = 5$. By the triangle inequality, max $|z| = |C| + 5 = 10$, so max $|z|^2 = 100$. 06 LEARNING INTELLIGENCE Learning intelligence Published revision priorities Priority revision register PRIORITY TOPIC CLASSIFICATION TIMEFRAME Kinematics Graphs High This week Vertical Motion Under Gravity High This week Motion Under Gravity High This week Consecutive Time Distances High This week Topic-level accuracy TOPIC SUBJECT ACCURACY LEVEL Kinematics Graphs Physics 0% Critical Vertical Motion Under Gravity Physics 0% Critical Acceleration Calculation Physics 100% Strong Motion Under Gravity Physics 50% Weak Acceleration-Time Graph Physics 100% Strong Consecutive Time Distances Physics 0% Critical Distance in nth Second Physics 100% Strong Multi-stage Motion Physics 0% Critical Instantaneous Rest Physics 0% Critical Uniform Acceleration Equations Physics 100% Strong Three-stage Motion Physics 0% Critical Newton's Second Law Physics 0% Critical Contact Forces Physics 0% Critical Pulley Systems Physics 0% Critical Friction and Pseudo Force Physics 100% Strong Tension and Breaking Strength Physics 0% Critical Equilibrium of Concurrent Forces Physics 0% Critical Connected Bodies and Pulleys Physics 0% Critical Topic-level accuracy · continued TOPIC

Mathematics · Q75 · explanation continued

SUBJECT ACCURACY LEVEL Relative Velocity Physics 0% Critical Uniform Motion Physics 0% Critical Variable Acceleration Physics 0% Critical Work-Energy Theorem Physics 0% Critical Relative Speed Physics 100% Strong Discovery of Subatomic Particles Chemistry 0% Critical Orbital Energies and Penetration Chemistry 0% Critical Electronic Configuration Chemistry 100% Strong Nodes in Orbitals Chemistry 100% Strong Photoelectric Effect and Ionization Chemistry 0% Critical Hydrogen Spectral Series Chemistry 0% Critical Aufbau Principle Chemistry 100% Strong Transition Metal Electronic Configuration Chemistry 0% Critical Bohr Model Transitions Chemistry 0% Critical Electromagnetic Spectrum Chemistry 0% Critical Photoelectric Effect Chemistry 0% Critical Excited State Configurations Chemistry 0% Critical de Broglie Wavelength Chemistry 100% Strong Valence Shell Configuration Chemistry 100% Strong Quantum Numbers Chemistry 100% Strong Orbital Symmetry and Waves Chemistry 100% Strong Bohr Radius Chemistry 100% Strong Photon Energy Chemistry 0% Critical Electron Gain Enthalpy Chemistry 0% Critical Radial Probability Distribution Chemistry 0% Critical Atomic Orbitals Chemistry 100% Strong Periodic Table Periods Chemistry 100% Strong Magnetic Moment Chemistry 100% Strong Bohr Energy Levels Chemistry 0% Critical Topic-level accuracy · continued TOPIC SUBJECT ACCURACY LEVEL Quantum Numbers Counting Chemistry 0% Critical Fractional Part Equations Mathematics 0% Critical Telescoping Series Mathematics 0% Critical Trigonometric Identities Mathematics 100% Strong Exponential Equations Mathematics 100% Strong Modulus Properties Mathematics 0% Critical Triangle Inequality Mathematics 0% Critical Telescoping Sums Mathematics 100% Strong Quadratic Inequalities Mathematics 0% Critical Newton's Theorem Mathematics 0% Critical Cosine Series Mathematics 0% Critical Euler Form Mathematics 0% Critical Rational Functions Range Mathematics 0% Critical Logarithmic Inequalities Mathematics 0% Critical Power Sets Mathematics 0% Critical Relations Between Roots Mathematics 100% Strong Harmonic Progression Mathematics 0% Critical Trigonometric Equations Mathematics 0% Critical Roots of Unity Mathematics 0% Critical Binomial / Generating Functions Mathematics 0% Critical Composite Quadratic Functions Mathematics 0% Critical Cube Roots of Unity & Probability Mathematics 0% Critical Geometric Progression Mathematics 0% Critical Location of Roots Mathematics 0% Critical Sine Product Identity Mathematics 100% Strong Geometric Locus & Circles Mathematics 0% Critical Mistake evidence QUESTION SOURCE DETAIL Q1 Unit Error · Kinematics Graphs Q2 Unit Error · Kinematics Graphs Q3 Formula Error · Vertical Motion Under Gravity Q7 Formula Error · Consecutive Time Distances Q9 Knowledge Gap · Multi-stage Motion Q10 Formula Error · Instantaneous Rest Q12 Unit Error · Three-stage Motion Q13 Unit Error · Newton's Second Law Q14 Calculation Error · Contact Forces Q15 Formula Error · Motion Under Gravity Q18 Calculation Error · Tension and Breaking Strength Q23 Formula Error · Variable Acceleration Q26 Calculation Error · Discovery of Subatomic Particles Q27 Knowledge Gap · Orbital Energies and Penetration Q30 Knowledge Gap · Photoelectric Effect and Ionization Q33 Calculation Error · Transition Metal Electronic Configuration Q35 Calculation Error · Electromagnetic Spectrum Q37 Calculation Error · Excited State Configurations Q44 Unit Error · Electron Gain Enthalpy Q55 Calculation Error · Modulus Properties Q60 Calculation Error · Cosine Series Q61 Incomplete Reasoning · Euler Form Q63 Formula Error · Logarithmic Inequalities Q66 Calculation Error · Harmonic Progression Q67 Calculation Error · Trigonometric Equations Q68 Calculation Error · Roots of Unity Weak chapters NO. WEAK CHAPTER 1 Kinematics 2 Laws of Motion 3 Structure of Atom 4 Sets and Relations 5 Sequences and Series 6 Trigonometry 7 Quadratic Equations 8 Complex Numbers 9 Inequalities Subject-wise revision sections Priority topics grouped by subject PRIORITY TOPIC CHAPTER ACCURACY CLASSIFICATION Physics Kinematics Graphs Kinematics 0% Critical Vertical Motion Under Gravity Kinematics 0% Critical Consecutive Time Distances Kinematics 0% Critical Multi-stage Motion Kinematics 0% Critical Instantaneous Rest Kinematics 0% Critical

Mathematics · Q75 · explanation continued

Three-stage Motion Kinematics 0% Critical Newton's Second Law Laws of Motion 0% Critical Contact Forces Laws of Motion 0% Critical Pulley Systems Laws of Motion 0% Critical Tension and Breaking Strength Laws of Motion 0% Critical Equilibrium of Concurrent Forces Laws of Motion 0% Critical Connected Bodies and Pulleys Laws of Motion 0% Critical Relative Velocity Kinematics 0% Critical Uniform Motion Kinematics 0% Critical Variable Acceleration Kinematics 0% Critical Work-Energy Theorem Laws of Motion 0% Critical Motion Under Gravity Kinematics 50% Weak Priority topics grouped by subject · continued PRIORITY TOPIC CHAPTER ACCURACY CLASSIFICATION Chemistry Discovery of Subatomic Particles Structure of Atom 0% Critical Orbital Energies and Penetration Structure of Atom 0% Critical Photoelectric Effect and Ionization Structure of Atom 0% Critical Hydrogen Spectral Series Structure of Atom 0% Critical Transition Metal Electronic Configuration Structure of Atom 0% Critical Bohr Model Transitions Structure of Atom 0% Critical Electromagnetic Spectrum Structure of Atom 0% Critical Photoelectric Effect Structure of Atom 0% Critical Excited State Configurations Structure of Atom 0% Critical Photon Energy Structure of Atom 0% Critical Electron Gain Enthalpy Classification of Elements 0% Critical Radial Probability Distribution Structure of Atom 0% Critical Bohr Energy Levels Structure of Atom 0% Critical Quantum Numbers Counting Structure of Atom 0% Critical Mathematics Fractional Part Equations Sets and Relations 0% Critical Telescoping Series Sequences and Series 0% Critical Modulus Properties Complex Numbers 0% Critical Triangle Inequality Complex Numbers 0% Critical Quadratic Inequalities Inequalities 0% Critical Newton's Theorem Quadratic Equations 0% Critical Cosine Series Trigonometry 0% Critical Euler Form Complex Numbers 0% Critical Rational Functions Range Quadratic Equations 0% Critical Logarithmic Inequalities Inequalities 0% Critical Power Sets Sets and Relations 0% Critical Harmonic Progression Sequences and Series 0% Critical Trigonometric Equations Trigonometry 0% Critical Roots of Unity Complex Numbers 0% Critical Binomial / Generating Functions Sequences and Series 0% Critical Composite Quadratic Functions Quadratic Equations 0% Critical Priority topics grouped by subject · continued PRIORITY TOPIC CHAPTER ACCURACY CLASSIFICATION Cube Roots of Unity & Probability Complex Numbers 0% Critical Mathematics Geometric Progression Sequences and Series 0% Critical Location of Roots Quadratic Equations 0% Critical Geometric Locus & Circles Complex Numbers 0% Critical Recommended next actions Weak topics · published list NO. WEAK TOPIC 1 Kinematics Graphs 2 Vertical Motion Under Gravity 3 Motion Under Gravity 4 Consecutive Time Distances 5 Multi-stage Motion 6 Instantaneous Rest 7 Three-stage Motion 8 Newton's Second Law 9 Contact Forces 10 Pulley Systems 11 Tension and Breaking Strength 12 Equilibrium of Concurrent Forces 13 Connected Bodies and Pulleys 14 Relative Velocity 15 Uniform Motion 16 Variable Acceleration 17 Work-Energy Theorem 18 Discovery of Subatomic Particles 19 Orbital Energies and Penetration 20 Photoelectric Effect and Ionization 21 Hydrogen Spectral Series Weak topics · published list · continued NO. WEAK TOPIC 22 Transition Metal Electronic Configuration 23 Bohr Model Transitions 24 Electromagnetic Spectrum 25 Photoelectric Effect 26 Excited State Configurations 27 Photon Energy 28 Electron Gain Enthalpy 29 Radial Probability Distribution 30 Bohr Energy Levels 31 Quantum Numbers Counting 32 Fractional Part Equations 33 Telescoping Series 34 Modulus Properties 35 Triangle Inequality 36 Quadratic Inequalities 37 Newton's Theorem 38 Cosine Series 39 Euler Form 40 Rational Functions Range 41 Logarithmic Inequalities 42 Power Sets 43 Harmonic Progression 44 Trigonometric Equations 45 Roots of Unity 46 Binomial / Generating Functions 47 Composite Quadratic Functions 48 Cube Roots of Unity & Probability 49 Geometric Progression 50 Location of Roots Weak topics · published list · continued NO. WEAK TOPIC 51 Geometric Locus & Circles MediXO EduX · Examination Intelligence End of report · 03 Nov 2026

05 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Mistake evidence

QUESTION	SOURCE DETAIL
Q1	Unit Error · Kinematics Graphs
Q2	Unit Error · Kinematics Graphs
Q3	Formula Error · Vertical Motion Under Gravity
Q7	Formula Error · Consecutive Time Distances
Q9	Knowledge Gap · Multi-stage Motion
Q10	Formula Error · Instantaneous Rest
Q12	Unit Error · Three-stage Motion
Q13	Unit Error · Newton's Second Law
Q14	Calculation Error · Contact Forces
Q15	Formula Error · Motion Under Gravity
Q18	Calculation Error · Tension and Breaking Strength
Q23	Formula Error · Variable Acceleration
Q26	Calculation Error · Discovery of Subatomic Particles
Q27	Knowledge Gap · Orbital Energies and Penetration
Q30	Knowledge Gap · Photoelectric Effect and Ionization
Q33	Calculation Error · Transition Metal Electronic Configuration
Q35	Calculation Error · Electromagnetic Spectrum
Q37	Calculation Error · Excited State Configurations
Q44	Unit Error · Electron Gain Enthalpy
Q55	Calculation Error · Modulus Properties
Q60	Calculation Error · Cosine Series
Q61	Incomplete Reasoning · Euler Form
Q63	Formula Error · Logarithmic Inequalities
Q66	Calculation Error · Harmonic Progression

Mistake evidence · continued

QUESTION	SOURCE DETAIL
Q67	Calculation Error · Trigonometric Equations
Q68	Calculation Error · Roots of Unity

Subject-wise revision sections

Recommended next actions

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